

Autodesk® Scaleform®

Scaleform

DirectX 9

Scaleform

3D

.

: Ben Mowery

: 3.09

: 2013 10 7

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Autodesk® Scaleform® 4.4

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1 소개

Scaleform

(UI)

GPU

UI

. Scaleform

UI

UI

Scaleform

3D

3D

3

3D

UI

, 3D

UI

, Scaleform

UI

Scaleform

UI

DirectX ShadowVolume SDK

주의: Scaleform

. Scaleform

Scaleform

Scaleform

Scaleform

주:

Scaleform

Scaleform

4

주:

DirectX

SDK ShadowVolume

"Cannot create renderer"

Scaleform Player

2 문서 개요

Scaleform Scaleform
(<http://gameware.autodesk.com/scaleform/developer/>)

- Scaleform SDK
- PDF
- : Scaleform
C++ API
- XML : SCALEFORM XML
- Scale9Grid: Scale9Grid
- IME : Scaleform IME
IME
- : Scaleform

3 설치와 빌드 의존성

3.1 설치

Windows Scaleform .
Scaleform DirectX . Scaleform
C:\Program Files\Scaleform\GFx SDK 4.4 .

3rdParty

libjpeg zlib Scaleform

Projects

Apps\Samples

Scaleform Player

Apps\Tutorial

Bin

FxPlayer: HUD

Samples: FLA , , ,

Win32: Scaleform Player 2

AMP : AMP

GFxExport: GFxExport

. 7

주: demo.sln

Win32

Include

Scaleform

Lib

Scaleform

Resources

CLIK

Doc

2

PDF

3.2 데모 컴파일

Scaleform

[→

→ Scaleform → GfX SDK 4.4 → Demo MSVC 9.0 Solutions → GfX 4.4 Demos.sln
D3D9_Debug_Static

Scaleform Player

C:\Program Files\Scaleform\GfX SDK

4.4\Bin\Win32\Msvc90\GfXPlayer\GfXPlayer_D3D9_Debug_Static.exe

[→

→ Scaleform → GfX SDK 4.4 → Demo Examples]

Scaleform Player D3Dx

SWF

Scaleform

3.3 Scaleform Player 에서 SWF 재생 벤치마크

→ Scaleform → GfX SDK 4.4 → GfX Players → GfX Player D3D9.
C:\Program Files\Scaleform\GfX SDK 4.4\Bin\Data\AS2\Samples\SWFToTexture
3D Windows.swf .



그림 1:

F1

1. CTRL+F . FPS

2. CTRL+W .

CTRL+W .

3. CTRL

4. CTRL

5. CTRL+Z .

6. CTRL+U

7. F2

800x600

1600x1200

Scaleform

Scaleform
Scaleform

"3D Scaleform Logo"

CTRL+W

. CTRL+A

. Edge Anti-Aliasing

(EdgeAA) ,

AA

(FSAA)

EdgeAA

draw primitive (DP)

EdgeAA

Scaleform Player

CPU

. CPU

Scaleform

. CTRL+Y

(60)

. CPU

주:

Scaleform

3.4 샘플 베이스 컴파일

Scaleform 4.4	Scaleform\GFx SDK 4.4\Apps\Tutorial	Tutorial Solution
.	Scaleform→ GFx SDK 4.4→ Tutorial	2005
2008	"Debug"	



그림 2: UI ShadowVolume

3.5 Scaleform 빌드 의존성

Scaleform			
\$(GFXDSK)	SDK		
C:\Program Files\Scaleform\GFx SDK 4.4\			
	\$(GFKSDK)		
"Msvc90"	2008	2010	Msvc90
Msvc10			

Scaleform	include	.
\$(GFXSDK)\Src		
\$(GFXSDK)\Include		
	"Additional Include Directories"	.
(GFx.h)	, AS3	AS3
"GFx/AS3/AS3_Global.h"		.
AS3		Scaleform LITE
.		

The following library directories should be added to the "Additional Library Directories" field for all build configurations:

```
"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\pcre\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\zlib-1.2.7\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\libpng-1.5.13\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\curl-7.29.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\jpeg-8d\lib\$(PlatformName)\Msvc90\Debug";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\$(ConfigurationName)"
```

주: Msvc90

Scaleform

libgfx.lib

libgfx_as2.lib

libgfx_as3.lib

libgfx_air.lib

libgfxexpat.lib

libjpeg.lib

zlib.lib

libcurl.lib

libpng.lib

libgfxrender_d3d9.lib

libgfxsound_fmod.lib

, Scaleform include . vcproj

Tutorial/Section3.5

3.5.1 AS3_Global.h 파일과 ObjWAS3_Obj_Global.xxx 파일

AS3_Global.h ObjWAS3_Obj_global.xxx 서로 연관성이 전혀 없는

AS3_Obj_Global.xxx " " ActionScript 3
swf " " () . C++
GlobalObjectCPP . Scaleform VM

AS3_Global.h . ClassRegistrationTable
AS3 C++

ClassRegistrationTable

AS3_Global.h

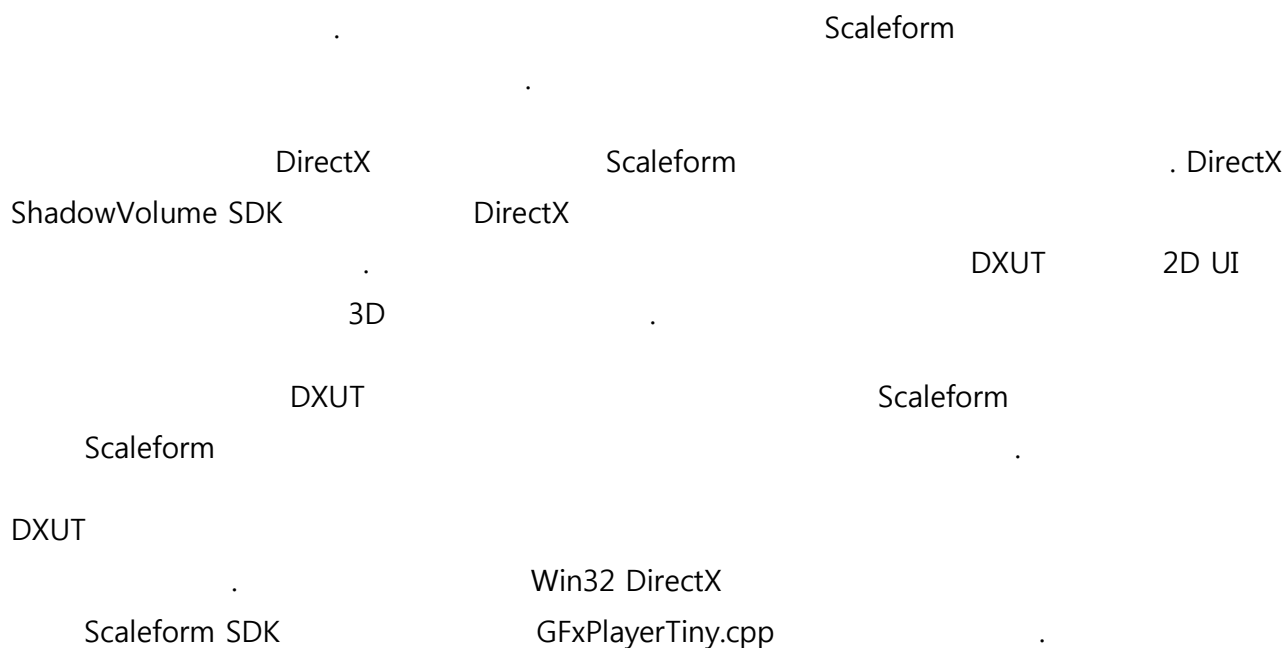
ClassRegistrationTable ()
ClassRegistrationTable include
AS3_Global.h

. AS3 VM

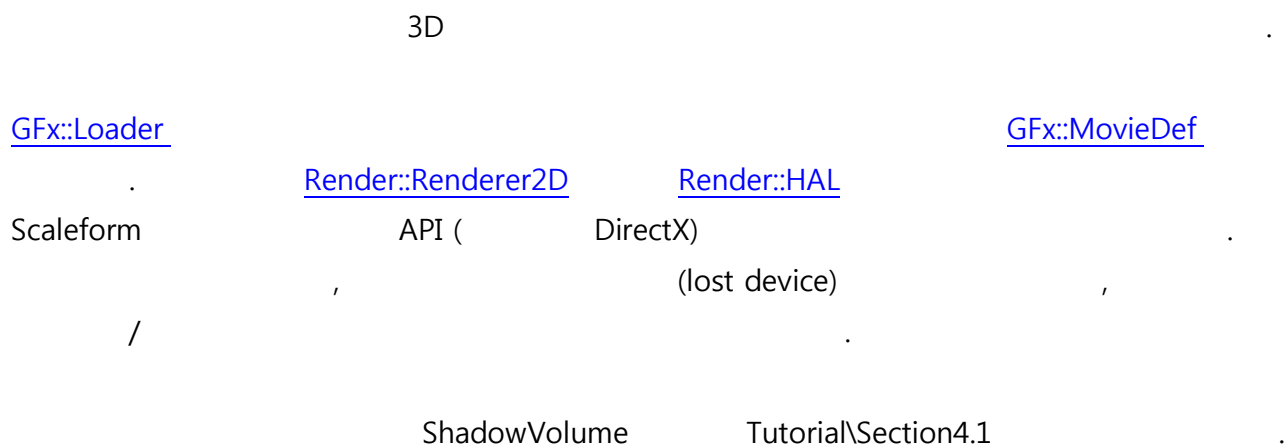
" "

4 게임 엔진 통합

Scaleform Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™ Trinigy Vision Engine™ 3D



4.1 플래시 렌더링



1 단계: 헤더 파일 추가

ShadowVolume.cpp

```
#include "GFx_Kernel.h"
#include "GFx.h"
#include "GFx_Renderer_D3D9.h"
```

Scaleform

GFxTutorial

Scaleform

Scaleform

```
// One GFx::Loader per application
Loader          gfxLoader;

// One GFx::MovieDef per SWF/GFx file
Ptr<MovieDef>    pUIMovieDef;

// One GFx::Movie per playing instance of movie
Ptr<Movie>       pUIMovie;

// Renderer data
Ptr<Render::D3D9::HAL> pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle    hMovieDisplay;
```

2 단계: GFx::System 초기화

Scaleform

Scaleform

[GFx::System](#)

. WinMain

```
// One GFx::System per application
GFx::System    gfxInit;
```

GFx::System

Scaleform

Scaleform

WinMain

GFx::System

Scaleform

GFx::System

Gfx::System

GfxTutorial

, Gfx::System::Init() Gfx::System::Destroy()

Gfx::System

3 단계: 로더 및 렌더러 생성

Scaleform

InitApp()

WinMain

InitApp()

```
gfx = new GfxTutorial();  
assert(gfx != NULL);  
if(!gfx->InitGfx())  
    assert(0);
```

GfxTutorial [Gfx::Loader](#)

Gfx::Loader

SWF/GFx

SWF/GFx

. Gfx::Loader

[Gfx::Log](#)

GfxTutorial::InitGfx()

Gfx::Loader

Gfx::Loader [SetLog](#)

Scaleform

Scaleform PlayerLog

Gfx::Log

```
// Initialize logging -- Scaleform will print errors to the log  
// stream.  
gfxLoader->SetLog(Ptr<Log>(*new Log()));
```

Gfx::Loader [Gfx::FileOpener](#)

Gfx::FileOpener

```
// Give the loader the default file opener
```

```
Ptr<FileOpener> pfileOpener = *new FileOpener;  
gfxLoader
```

5 단계: 무비 인스턴스 생성

```
GFx::MovieDef      GFx::Movie
GFx::Movie
,
,
.
.

pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.GetPtr() != NULL);
```

[CreateInstance](#)

```
false
(Arena)
```

[Memory System Overview](#)

```
Advance()
false CreateInstance
.

// Advance the movie to the first frame
pUIMovie->Advance(0.0f, 0, true);

// Note the time to determine the amount of time elapsed between
// this frame and the next
MovieLastTime = timeGetTime();
```

[Advance](#)

```
3D
pUIMovie->SetBackgroundAlpha(0.0f);
```

3D

6 단계: 디바이스 초기화

DirectX

Scaleform

Render::D3D::HALInitParams

```

Render::D3D9::HAL::InitHAL          , D3D          ,
Scaleform          . D3D
InitHAL          /
.

ShadowVolume  OnResetDevice          DXUT          ,
.          GfxTutorial

OnResetDevice          .

pRenderHAL->InitHAL(
    Render::D3D9::HALInitParams(pd3dDevice, presentParams,
        Render::D3D9::HALConfig_NoSceneCalls));

InitHAL          Scaleform  D3D          .
HAL_NoSceneCalls          DirectX BeginScene()  EndScene()          Scaleform
.          ShadowVolume          OnFrameRenderer
.

```

7 단계: 로스트 디바이스

```

.          , D3D
.          D3D
. ShadowVolume  OnLostDevice          . Render::HAL
.          GfxTutorial          OnLostDevice          D3D
.

pRendererHAL->ShutdownHAL();

DXUT
.          Win32/DirectX
Scaleform SDK          GfxPlayerTiny.cpp          .

```

8 단계: 자원 할당 및 청소(cleanup)

```

Scaleform          GfxTutorial          WinMain
GfxTutorial          .

delete gfx;
gfx = NULL;

```

```

DirectX 9 . Scaleform
InitHAL()

ShutdownHAL()

DirectX9 InitHAL D3DPOOL_DEFAULT
D3DPOOL_DEFAULT

InitHAL

ShutdownHAL D3DPOOL_DEFAULT . ShadowVolume
DXUT DXUT OnResetDevice
D3DPOOL_DEFAULT OnLostDevice
GFxTutorial::OnLostDevice ShutdownHAL GFxTutorial::OnResetDevice
InitHAL

D3DPOOL_DEFAULT
InitHAL ShutdownHAL

```

9 단계: 뷰포트 설정

```

D3D
GFxTutorial::OnResetDevice

// Use the window client rect size as the viewport.
RECT windowRect = DXUTGetWindowClientRect();

```

Scaleform

4.2

10 단계: DirectX 썬으로 렌더링

```
ShadowVolume OnFrameRender() . D3D
    BeginScene() EndScene() . EndScene()
    GfxTutorial::AdvanceAndRender() .
    void AdvanceAndRender(void))
```

DX9

GfxTutorial::AdvanceAndRender()

```
// Save DirectX state before calling Scaleform  
g_pStateBlock->Capture();  
// Render the frame and advance the time counter  
gfx->AdvanceAndRender();  
// Restore DirectX state to avoid disturbing game render state  
g_pStateBlock->Apply();
```

12 단계: 기본 UI 사용막기

DXUT

UI

Section4.1\ShadowVolume.cpp

DXUT

```
// Disable default UI
```

..

DirectX



그림 3: Scaleform

UI

ShadowVolume

4.2 크기조절 모드

Gfx::Movie::SetViewport

Scaleform

•

•

4:3

Scaleform

1600x1200

). Scaleform

(, 800x600

[Gfx::Movie::SetViewScaleMode](#)

GfxTutorial::InitGfx()

Gfx::Movie

pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);

SetViewScaleMode

SM_NoScale	
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

[SetViewAlignment](#)

SM_NoScale

SM_ShowAll

SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment SetViewScaleMode SM_NoScale

Scaleform

d3d9guide fla

C++

. SetViewScaleMode

SetViewAlignment

Stage

(Stage.scaleMode, Stage.align)

4.3 입력 이벤트 처리

ShadowVolume

Scaleform

The [Gfx::Movie::HandleEvent](#)

Gfx::Event

Gfx::Movie

4.3.1 마우스 이벤트

ShadowVolume

MsgProc

Win32

Scaleform

GfxTutorial::ProcessEvent

WM_MOUSEMOVE, WM_LBUTTONDOWN

WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);
    if (pUIMovie)
    {
        if (uMsg == WM_MOUSEMOVE)
        {
            MouseEvent mevent(Gfx::Event::MouseMove, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONDOWN)
        {
            ::SetCapture(hWnd);
            MouseEvent mevent(Gfx::Event::MouseDown, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONUP)
        {
            ::ReleaseCapture();
            MouseEvent mevent(Gfx::Event::MouseUp, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
    }
}
```

Scaleform

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,  
                    screen_width, screen_height, 0);
```

(0, 0)

#2:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,  
                    screen_width / 4, screen_height / 4, 0);
```

HandleEvent

#3:

```
movie_width = screen_width / 6;  
movie_height = screen_height / 6;  
pMovie->SetViewport( screen_width, screen_height,  
                    screen_width / 2 - movie_width / 2,  
                    screen_height / 2 - movie_height / 2,  
                    movie_width, movie_height);
```

(0, 0)

(screen_width /

2 - movie_width / 2, screen_height / 2 - movie_height / 2)

[Gfx::Movie::SetViewAlignment](#)

[Gfx::Movie::SetViewport](#)

SetViewAlignment

[SetViewScaleMode](#)

Scaleform

4.3.2 키보드 이벤트

[Gfx::Movie::HandleEvent](#)

[. Gfx::KeyEvent](#)

Gfx::CharEvent

```
KeyEvent(EventType eventType = None,
          Key::Code code = Key::None,
          UByte asciiCode = 0,
          UInt32 wcharCode = 0,
          UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

Gfx::KeyEvent

, Gfx::CharEvent

ASCII

Gfx::CharEvent WM_CHAR

. Gfx::KeyEvent

WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP

- SHIFT 'c' : WM_KEYDOWN

Gfx::KeyEvent

- 'c'

-

- SHIFT

- " " ASCII 'C' Scaleform WM_CHAR

Gfx::CharEvent

- 'c' WM_KEYUP

Gfx::KeyEvent

Gfx::CharEvent

- F5 : Gfx::KeyEvent

. F5

ASCII

Gfx::CharEvent

Gfx::KeyEvent

, Gfx_Event.h

. GfxPlayerTiny.cpp Scaleform Player

ProcessKeyEvent

```
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM lParam)
```

```
    WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP,
    WM_KEYDOWN, WM_KEYUP Scaleform
    pMovie

    Gfx::KeyEvents Gfx::CharEvents
    Gfx::CharEvents ASCII
    ( , IME) . IME ,
    ( IME
    ) Gfx::KeyEvent
    Page Up Page Down
```

Tutorial\Section4.3

```
    3D
    "Settings" C++ DX9
    d3d9guide fla
    "Change Mesh"
    "Change Mesh"
```

4.3.3 히트 테스트

UI

?

UI 3D

GFx::Movie::HitTest

GFxTutorial::ProcessEvent

[HitTest](#)

. UI

DXUT

```
bool processedMouseEvent = false;
if (uMsg == WM_MOUSEMOVE)
{
    MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONDOWN)
{
    ::SetCapture(hWnd);
    MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONUP)
{
    ::ReleaseCapture();
    MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}

if (processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
                                             Movie::HitTest_Shapes))
    *pbNoFurtherProcessing = true;
```

4.3.4 키보드 포커스

"Change Mesh"

4.3.2

W, S, A, D, Q, E

3D

Scaleform

3D

. 6.1.2

C++

4.3.5 터치 이벤트

```

// Scaleform
// Windows API
// Windows 7
// Windows
// WINVER=0x601
// Windows API
// ShadowVolume.cpp
#define ENABLE_MULTITOUCH

#include <windows.h>

RegisterTouchWindow
WM_TOUCH

MsgProc

if(uMsg == WM_TOUCH)
{
    ProcessTouchEvent(pUIMovie, uMsg, wParam, lParam);
}

ProcessTouchEvent WM_TOUCH

Windows
Gfx::TouchEvent

HandleEvent
Gfx::TouchEvent
ID(OS
), x/y
(0~1)

TouchEvent( EventType evtType,
            unsigned id,
            float _x,
            float _y,
            float wcontact = 0,
            float hcontact = 0,
            bool primary = true,
            float pressure = 1.0f)

SWF
Gfx 0
Gfx::GestureEvent

```

MultitouchInterface

Scaleform

```
class FxPlayerMultitouchInterface : public MultitouchInterface
{
public:
    //
    virtual unsigned GetMaxTouchPoints() const { return 2; }

    //
    virtual UInt32 GetSupportedGesturesMask() const { return 0; }

    //
    virtual bool SetMultitouchInputMode(MultitouchInputMode) { return
true; }
};
```

SetMultitouchInterface

5 지역화와 폰트 소개

5.1 폰트개요 및 기능

Scaleform

TTF

Scaleform

<http://gameware.autodesk.com/scaleform/developer/?action=doc>

5000

Scaleform

(glyph-packing)

Scaleform

. Scaleform

TTF

Scaleform

"fake italic"

"fake bold"

Bin\Data\AS2\Samples\FontConfig

sample.swf

Scaleform Player

D3D9

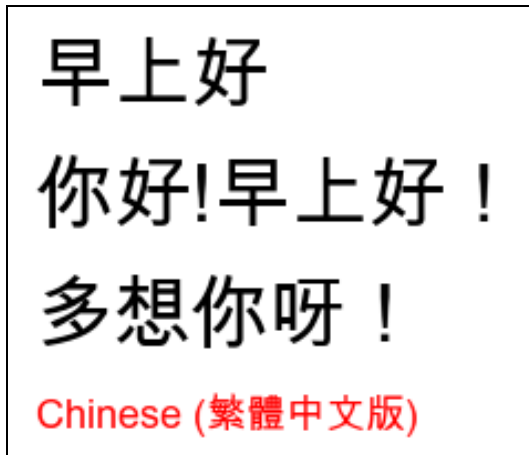


그림 4a:

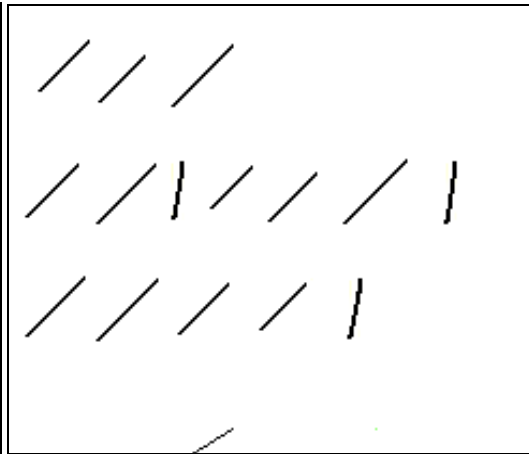


그림 4b:

CTRL+W

2



그림 4c:

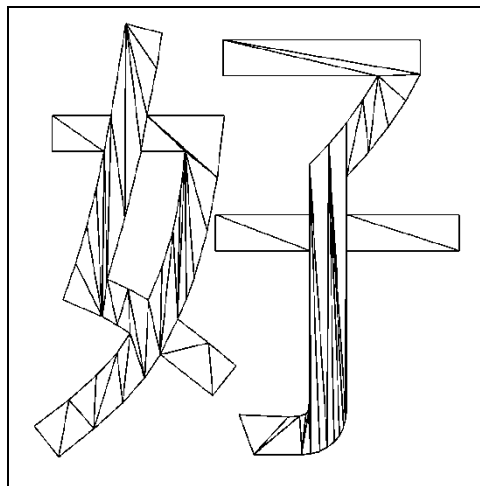


그림 4d:

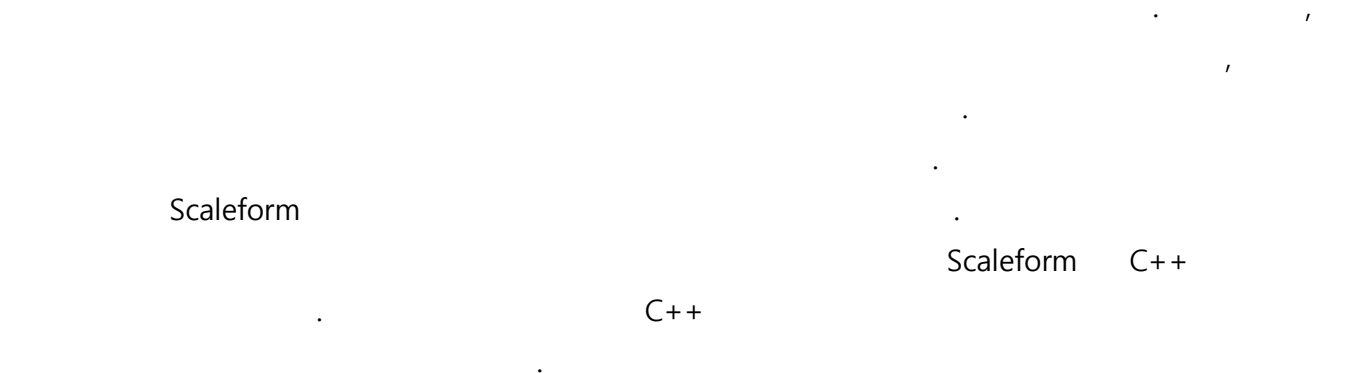
4c 4d , Scaleform

gfx_2.1_texteffects_sample.zip

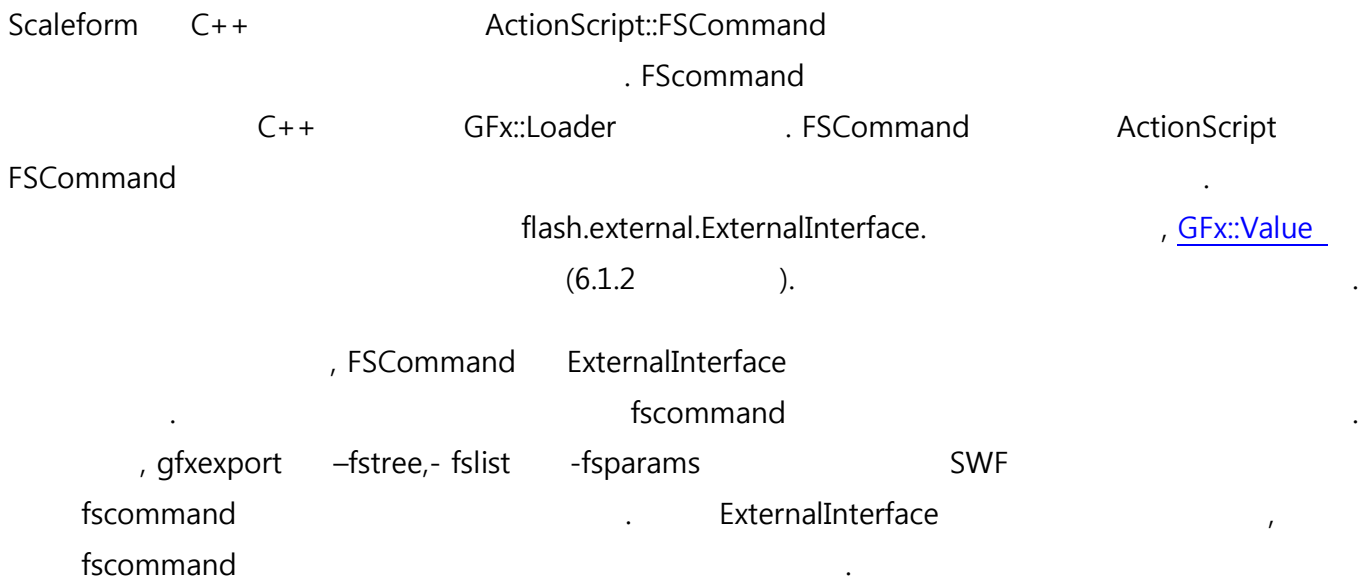


그림 5:

6 C++, 플래시, 액션스크립트 연결하기



6.1 액션스크립트에서 C++로



6.1.1 FS Command 콜백



```

        fscommand
        . ExternalInterface

        Gfx FSCommand
        .
Gfx::FSCommandHandler fscommand , Gfx::Loader
Gfx::Movie
        fscommand
        . GfxPlayerTiny
        ("FxPlayerFSCommandHandler" ) ShadowVolume
        Tutorial\Section6.1

, Gfx::FSComandHandler

class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                        const char* pcommand, const char* parg)
    {
        GfxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};

        fscommand
        fscommand

, GfxTutorial::InitGfx() Gfx::Loader

// Register our FSCommand handler
Ptr<GfxFSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);

Gfx::Loader
        Gfx::Movie
SetFSCommandHandler

        fscommand
        . ShadowVolume

        "Toggle Fullscreen"
        . UI

        FSCommand: ToggleFullscreen, Args:

```

Open Flash/HUDMgr.as in Flash Studio. This class is referenced from d3d9guideAS3.fla. The association between this external class and the Flash content is made by setting the ActionScript class for the "hudMgr" Symbol in d3d9guideAS3's Symbol Library.

Compare the events printed to the screen with the fscommand() calls made from the HUDMgr class in the following function:

```
function toggleFullscreen(ev:MouseEvent) {
    fscommand("ToggleFullscreen");
}
```

```
fscommand("ToggleFullscreen") fscommand("ToggleFullscreen", hud.visible)
hud.visible.value C++ . (CTRL+ALT+Shift+S)
d3d9guideAS3.swf .
```

```
UI FSCCommand ShadowVolume
. , fscommand
.
```

```
if (strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

```
DXUT OnFrameMove . OnFrameMove
.
```

```
if (doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

```
, .
Advance Invoke .
```

6.1.2 ExternalInterface

```
ExternalInterface.call fscommand
.
```

ExternalInterface

fscommand

```
class OurExternalInterfaceHandler : public ExternalInterface
{
public:
virtual void Callback(Movie* pmovieView,
                     const char* methodName,
                     const Value* args,
                     unsigned argCount)
{
    GfxPrintf("ExternalInterface: %s, %d args: ",
              methodName, argCount);
    for(unsigned i = 0; i < argCount; i++)
    {
        switch(args[i].GetType())
        {
        case Value::VT_Null:
            GfxPrintf("NULL");
            break;
        case Value::VT_Boolean:
            GfxPrintf("%s", args[i].GetBool() ? "true" : "false");
            break;
        case Value::VT_Int:
            GfxPrintf("%s", args[i].GetInt());
            break;
        case Value::VT_Number:
            GfxPrintf("%3.3f", args[i].GetNumber());
            break;
        case Value::VT_String:
            GfxPrintf("%s", args[i].GetString());
            break;
        default:
            GfxPrintf("unknown");
            break;
        }
        GfxPrintf("%s", (i == argCount - 1) ? "" : ", ");
    }
    GfxPrintf("\n");
}
};
```

Gfx::Loader

```
Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);
```

. Open d3d9HUD.as used by d3d9guideAS3.fla and look at the ActionScript for the following functions:

```
function onSubmit(path:String) {
    ExternalInterface.call("MeshPath", path);
    CloseMeshPath();
}
```

```

}

function onFocusIn(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", true);
}

function onFocusOut(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", false);
}

```

MeshPathFocus

```

ExternalInterface    ExternalInterface    ,    (    )
    ,    "MeshPathFocus"

```

```

Callback! MeshPathFocus, nargs = 1
    arg(0) = true

Callback! MeshPathFocus, nargs = 1
    arg(0) = false

```

GFxTutorial

```

if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
args[0].GetType() == Value::VT_Boolean) {
    if (args[0].GetType() == Value::VT_Boolean)
        gfx->SetTextboxFocus(args[0].GetBool());
}

```

GFxTutorial::ProcessEvent

3D

```

if (uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
    uMsg == WM_CHAR)
{
    if (textboxHasFocus || wParam == 32 || wParam == 9)
    {
        ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
        *pbNoFurtherProcessing = true;
    }
}

```


(ASCII 32) (ASCII 9) , "Toggle UI" "Settings"

"Change Mesh"

ExternalInterface

. ExternalInterfaceHandler::Callback

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
    doChangeMesh = true;
    const char *filename = args[0].GetString();
    MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
        changeMeshFilename, _countof(changeMeshFilename));
}
```

DXUT OnFrameMovie

DXUT

6.2 C++에서 액션스크립트로

6.1 C++ C++

Scaleform

Scaleform C++ (get) (set)

6.2.1 액션스크립트 변수 조작

Scaleform [GetVariable](#) [SetVariable](#)

. Tutorial\Section6.2

Scaleform Player

HUD

(fxplayer.swf) F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
```

```

        LPARAM lParam, bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);

    if (pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if (wParam == VK_F5)
        {
            int counter = (int)pHUDMovie
                ->GetVariableDouble("_root.counter");
            counter++;
            pHUDMovie->SetVariable("_root.counter",
                Value((double)counter));

            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_root.MessageText.text", str);
        }
    }
    ...
    GetVariableDouble _root.counter
    GetVariableDouble 0
    _root.counter . Gfx::Movie
    SetVariable
    GetVariable SetVariable

```

fxplayer

6 9

69 □



그림그림 8: HUD

SetVariable

SetVariable "sticky"

GFx::Movie::SetVarType

_root.mytextfield 3

SetVariable("_root.mytextfield.text", "testing", SV_Normal) 1

() SV_Sticky

_root.mytextfield.text , 3

C++

SV_Normal

SV_Sticky

SV_Normal

SetVariableArray

GFxTutorial::InitGFx() _root.SceneData

```
// Initialize the scene dropdown
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
```

`"_root.SceneData", 0, sceneData, 3);`

Tutorial\Section6.1

ExternalInterface

ShadowVolume

주:

C++

C++

6.2 , d3d9guide.swf/fla

6.2.2 액션스크립트 서브루틴 실행

[Gfx::Movie::Invoke](#)

UI

UI

Gfx::Movie::Invoke

F6

"Change Mesh"

F7

SetVariable

. SetVariable

Invoke

```

Invoke
    , Scaleform
    . 1
    . true
initFirstFrame Gfx::MovieDef::CreateInstance
    Invoke printf
    . Gfx::Value
    . InvokeArgs Invoke
    va_list . Invoke InvokeArgs
    printf vprintf
    .

```

6.3 플래시 파일간 통신

```

C++ . UI SWF
    , C++
    , (MMOG) HUD, HUD,
    .
    HUD HUD
    .
    SWF
    Gfx::Movie C++
    .
    loadMovie unloadMovie
    SWF Gfx::Movie . SWF Gfx::Movie
    , C++
    .
    ,
    container.swf
    .

```

MovieClipLoader

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

```
function loadMovie(url:String):void {  
    var loader = new Loader();  
    loader.load( new URLRequest( url ) );  
    addChild( loader )  
}
```

7 텍스처에 렌더링하기

3D

Flash Studio

SWF

SWF

UI

UI



1.

cell.x

(vertex)

(cellwall.jpg)

UI

cellwallRT.jpg

MeshMaterialList

```
Material {  
    1.000000;1.000000;1.000000;1.000000;;  
    40.000000;  
    1.000000;1.000000;1.000000;;  
    0.000000;0.000000;0.000000;;  
    TextureFilename {  
        "cellwallRT.jpg";  
    }  
}
```

2. cell.x DXUT

(parsing) . DXUT cell.x
(material)

CreateTexture

Render

Texture

DXSDK

DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,  
    D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,  
    D3DP00L_DEFAULT, &g_pRenderTex, 0);
```

```
g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;  
ptex->Release();
```

3. Gfx AdvanceAndRender

a. ,

```
pd3dDevice->GetRenderTarget(0, &polddSurface);  
pd3dDevice->GetDepthStencilSurface(&polddDepthSurface);
```

b. (surface)

```
ptex->GetSurfaceLevel(0, &psurface);  
HRESULT hr = pd3dDevice->SetRenderTarget(0, psurface );
```

c. Display a

8 OpenGL 샘플

Nvidia OpenGL/GLUT (TutorialWOpenGL)

(<http://developer.download.nvidia.com/SDK/10/opengl/samples.html#tessellation>).

D3D9 tessellation.cpp
 tessellation_original.cpp NVIDIA
 . Init(), ShutDown, AdvanceAndDisplay()
 Gfx Gfx GfxPlayerImpl (GfxPlayerGL.cpp)
 .

9 GfxExport 로 전처리

SWF . SWF ,
 SWF SWF .
 GfxExport SWF .
 ,
 DXT DDS
 .
 . GfxExport .
 Scaleform . GfxExport
 . d3d9guide.swf fxplayer.swf Scaleform .
 gfxexport -i DDS -c d3d9guideAS3.swf
 gfxexport -i DDS -c fxplayer.swf
 gfxexport.exe C:\Program Files\Scaleform\\$(GFXSDK)
 Tutorial\Section7 convert.bat .

	DDS	. DDS	DirectX
-i			
	DXT		4

–c . Scaleform .
DXT .

```
-share_images                               SWF
                                           .
```

Tutorial\Section8	ShadowVolume	Gfx::Loader::CreateMovie
.		

10 다음 단계

Scaleform

- [Scaleform Player SWF to Texture SDK](#),
Gamebryo™ Unreal® Engine 3
- [FAQs](#)
- [Scale9Grid Overview](#) Scale9
- : Scaleform SWF [Gfx::FileOpener](#)
- [Gfx::ImageCreator](#)
- [Gfx::Translator](#)